Introduction of YTTEK YTPC1002C01 Platform

YTTEK

Release Date: Oct/2025

Outline

- Introduction of YTPC-Series SDR Platform
- Computer Environment Setting
- Basic Function: TX/RX/Loopback
  - Example: Single Tone
- Communication Example:
  - SISO OFDM

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SDR Wireless Development Platform

PluSDR Lite™

Software-Defined Radio Platform

- Up to 100M of bandwidth
- Maximum supported sampling rate: 122.88 MHz
- Free sample code adaptive: Matlab, C++, Python
- Independent TX/RX for TDD/FDD full-duplex operation
- Small form factor

Application

Wireless | Beam-Forming | MIMO

Item	Spec.		Remark
Bandwidth	100		MHz
Frequency	300M-6G		Hz
MIMO	2T2R		
TX max output power	700 MHz 3.5 GHz 5.5 GHz	18 16 16	dBm
Power control range	50		dB
Power control resolution	0.25		dB
Max sampling frequency	122.88M		Hz
RX max input power	+10		dBm
Gain range	50		dB
Gain step	0.5		dB

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YTPC-Series Platform
API Introduction and Quick Start Manual

For MATLAB Development Environment

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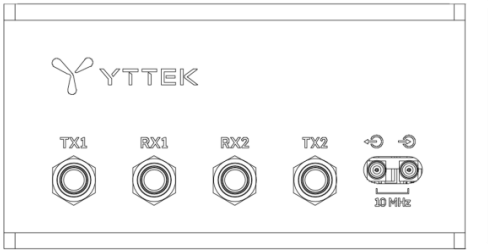
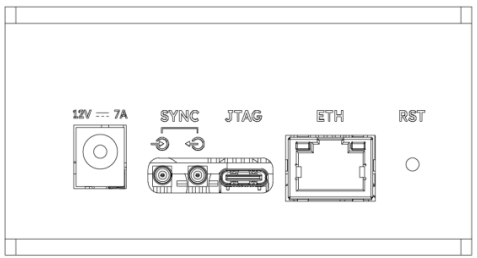
Kit Contents

FRONT



BACK



1. YTPC1002C01 platform
2. DC Power adapter 12V with 110V / 220V power cord
3. 1G Ethernet cable
4. SMA (Male)-to-SMA (Male) cable 2 pieces (option)

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Required Software

- MATLAB R2018a ~ R2023a
- Windows 10 (Win 64-bit)

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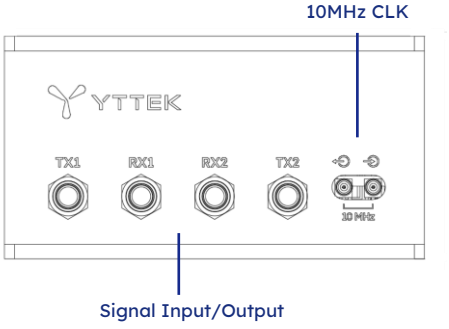
I/O Port Descriptions

FRONT



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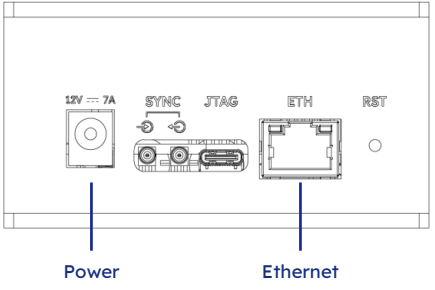

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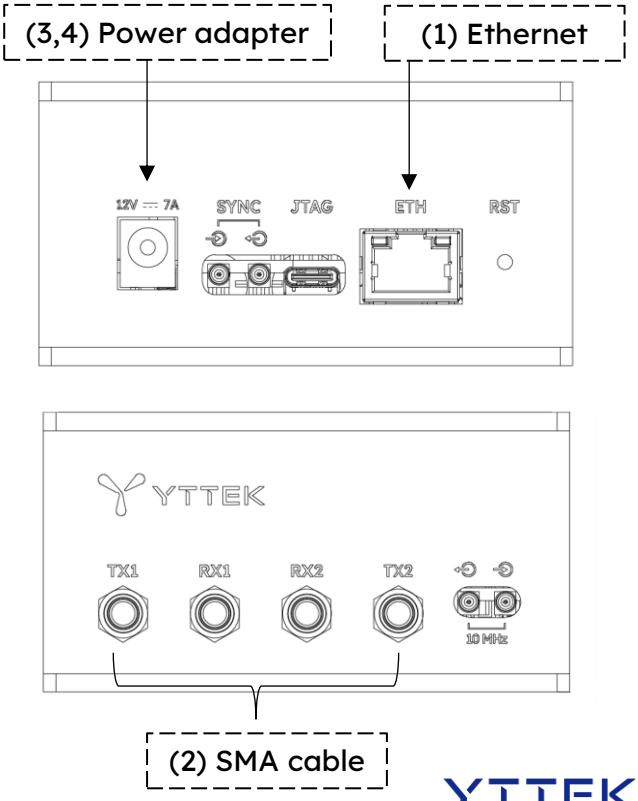
Hardware setup

1. Use the Ethernet cable to connect the YTPC1002C01 platform (ETH) to the computer.
2. Connect SMA cables to confirm signal source.
3. Connect the 110V AC to 12V DC power supply to the YTPC1002C01 platform.
4. Plug the 110V power cord of power supply into a wall outlet.



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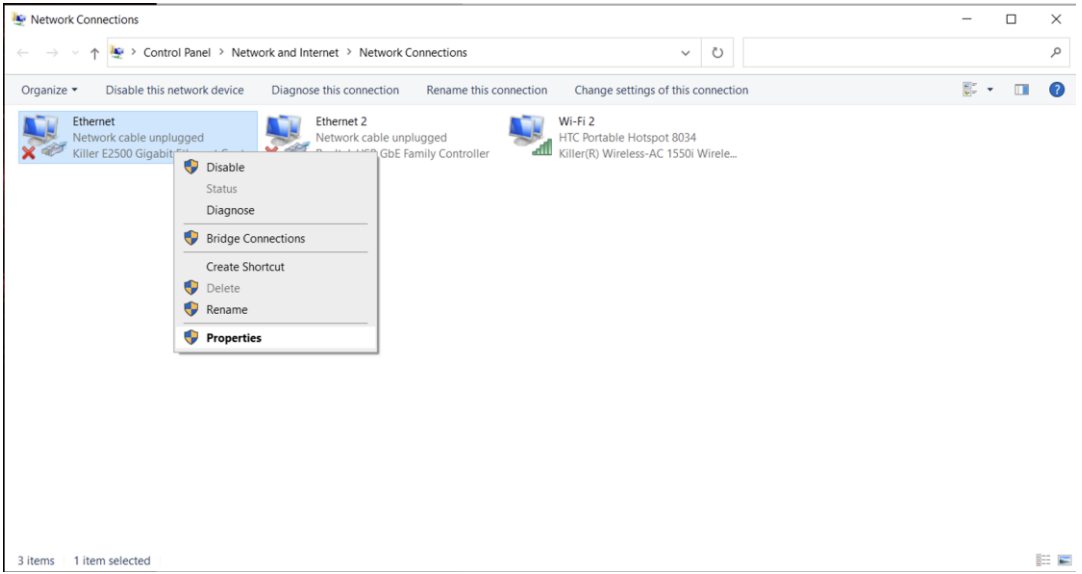
Computer Environment Setting: IP Address

- Setting a static IP address on your computer: **192.168.3.200**



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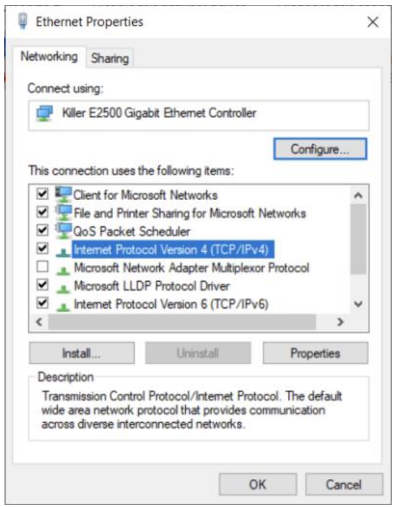
Set IP Address to 192.168.3.200.

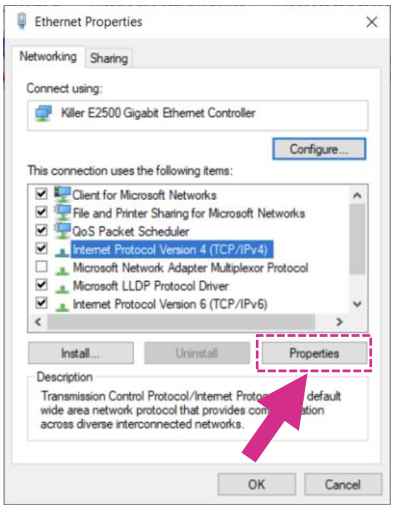
Step1

Tick of all the setting except TCP/IPv4.



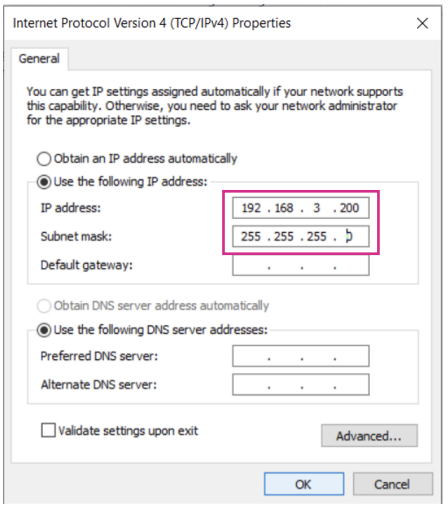
Step2

Choose “Properties”



Step3

Fill with the settings.



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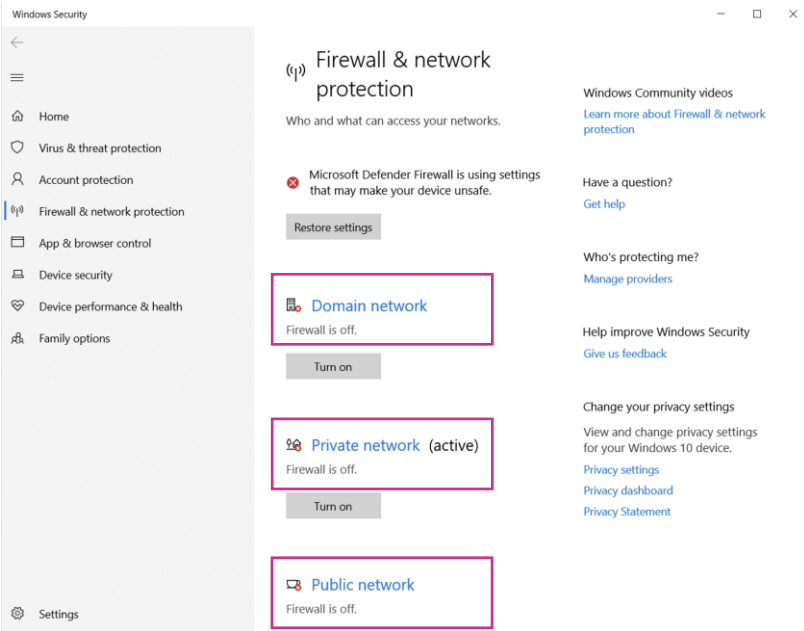
Firewall Setting: Turn off Firewall



Notes:
Please disable the firewall during use to prevent packets from being blocked.

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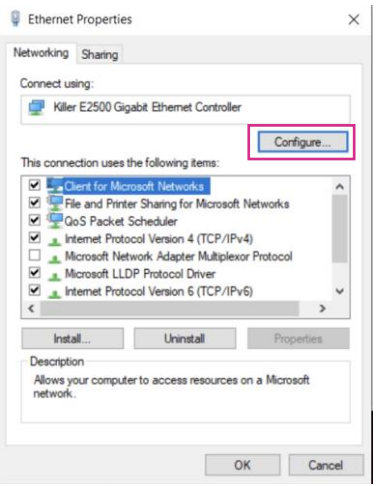
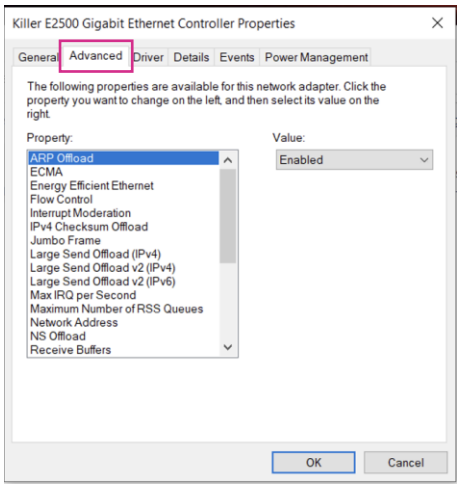
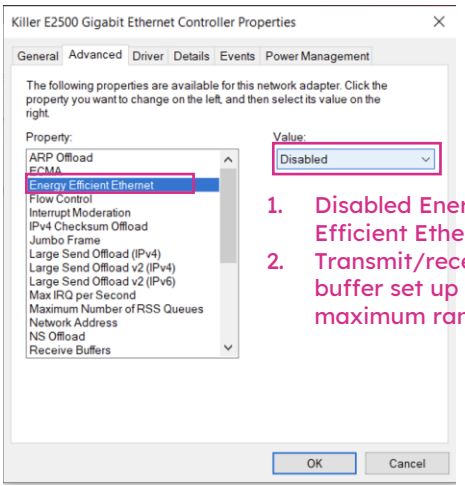
Ethernet advance setting

- For better performance, we recommend to set ethernet configure as follows:







1. Disabled Energy Efficient Ethernet.
2. Transmit/receive buffer set up to the maximum range.

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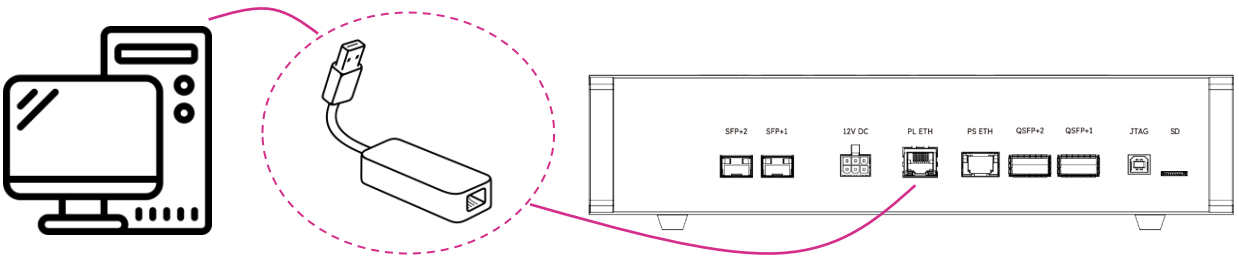
Ethernet Issue

- We recommend to connect ethernet directly to PC/laptop.
- If you want to use usb/ethernet adapter, we recommend to choose “Realtek chip or intel” USB 3.0 to gigabit Ethernet adapter.



USB 3.0 to gigabit Ethernet adapter

USB2.0

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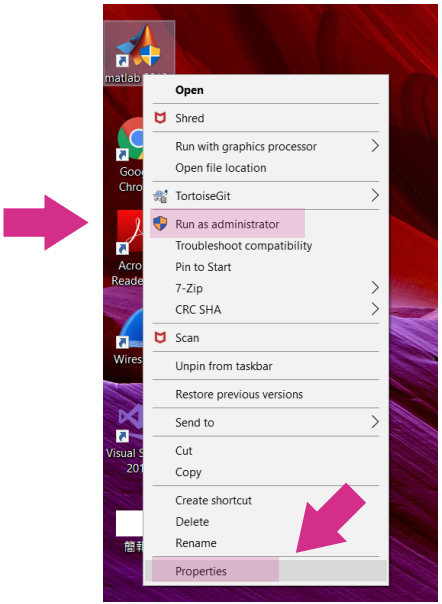
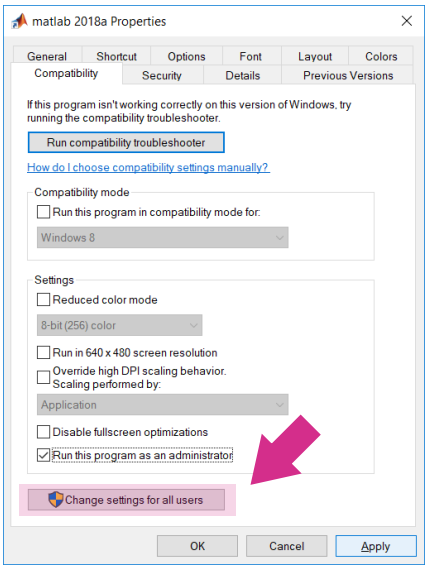
Run MATLAB

- Run as administrator





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API List

- **Init()**
  - Initialize platforms.
- **RX()**
  - Receive data from RF.
- **TX() / TX_start()**
  - Transmit data to RF.
- **TX_RX()**
  - Loopback function.
- **set_RX_Ref_Level_ELSDR()**
  - RX reference level setting
- **set_TX_power()**
  - TX absolute power setting (dBm).
- **LO_CHANGE()**
  - Change center frequency of YTPC1002C01.

- **Communication Examples**
  - Transmit data
  - Receive data
  - Loopback data
  - SISO OFDM
  - MIMO OFDM
  - OFDM with coding

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Basic functions

TX	• TX(xk, rep, time), TX_start()
RX	• rk=RX(N_RF_Chain, N_sample)
Loopback	• rk=TX_RX(xk, TX_REP, RX_LEN, RX_delay, LB_mode)
RF Setting	• Set Tx power, Rx Ref. Level, Change center frequency, ……

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Init () – Single Platform

- Init ()
  - The default IP address of platform is set to **192.168.3.100**.
  - The index of SDR is **0**.

```
>> Init
==========================================================================================
!!! Before system initialization, Please check one LED is ON only in each SDR hardware
Now, detecting exist SDR hardware ! Please wait...
==========================================================================================
 Software version: 2.10.2456710

 SDR idx 0: HW version: 8.3
 Initial active HW OK !!!
 clock set to 1500(mv), sdr 0
 Set Tx Attenuation OK !!!
 Set Rx Gain OK !!!
 ==========================================================
  Initial Completed !
 ==========================================================
```

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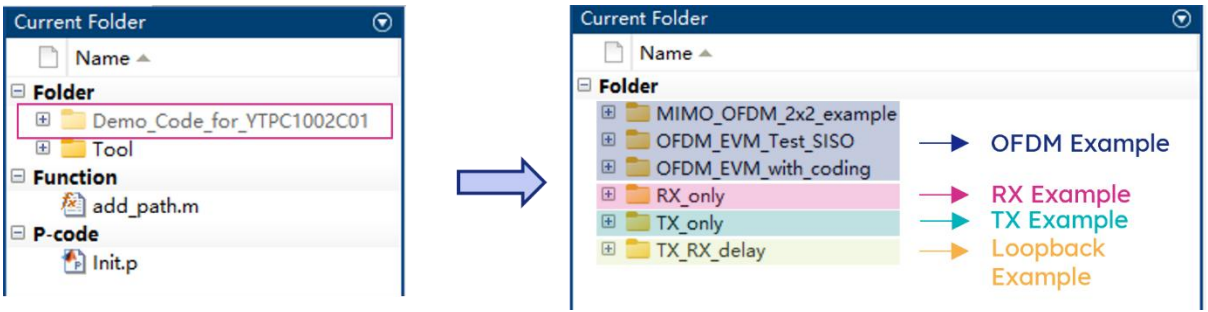
Run Example

- All YTPC1002C01 examples are in the folder ".\release\Demo_Code_for_YTPC1002C01".



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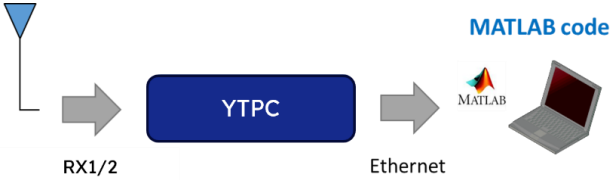
RX()



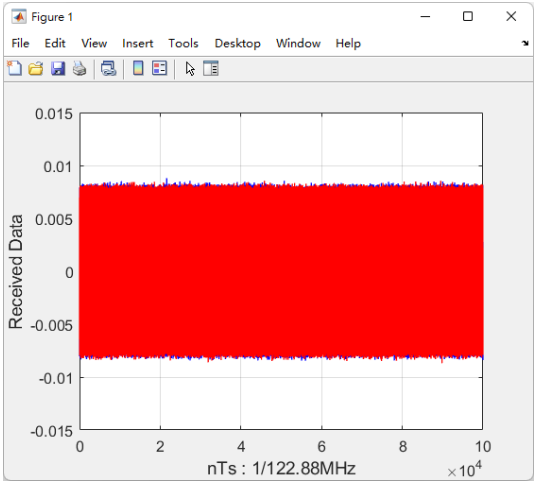
- RX(number_of_rows, length)
  - Use the number of rows in data to determine how many antennas to use to receive signals.
  - length: number of samples users want to receive.
  - One output parameter is needed.

```
Number_of_RX_rows=1;  % 1:SISO, 2:MIMO
rx_len=1e5;
Pow_com = set_RX_Ref_Level_ELSDR([0 10]);
rk = RX(Number_of_RX_rows,rx_len);
P_rk=10*log10(mean(abs(rk).^2))-Pow_com;
```

```
>> size(rk)

ans =

          1      100000
```



Note: The preferred length is below 1e8 samples. (Getting more received samples is dependent on the capability of PC).

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RX()

rk

$$\underbrace{\begin{bmatrix} 0.2+i0.5 & \cdots & 0.9-i0.1 \\ -0.3+i0.7 & \cdots & -0.2-i2 \end{bmatrix}}_{\text{Number of samples}} \left.\vphantom{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}\right\} \text{Dimension of RF chain}$$

- The range of the each real and imagery of each element of rk are $-1 \leq real(rk(m,n)) \leq 1$ and $-1 \leq imag(rk(m,n)) \leq 1$. ±1 means saturation.
- The time duration between $rk(m,n)$ and $rk(m,n+1)$ is the inverse of the sampling rate. In YTPC1002C01 the default sampling rate is 122.88MHz.
- *m*th row of *rk* corresponding to *m*th RX RF chain
- The limitation of total number of elements of rk is less than 1e8 (numel(rk) <= 1e8).
- Example code are in ".\release\ Demo_Code_for_YTPC1002C01\RX_only"
  - "rx_scope.m" : simple example for real time show rx signal
  - "rx_spectrum.m": simple example for spectrum by using MATLAB function pwelch()
  - "RX_with_RX_gain.m" and related function: simple example to show how to use RX()

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set_RX_Ref_Level_ELSDR()

- Pow_com = set_RX_Ref_Level_ELSDR()
  - Input array: ([RF_chain0 Amp_val0; RF_chain1 Amp_val1])
  - Pow_com: Compensation value returned by the function.
    - Represents the calibration offset of the selected RX Ref Level.
    - Required when converting relative RX power into absolute dBm.
- For example:
  - set RX Reference Level to -10 dBm
    - set_RX_Ref_Level([0 -10; 1 -10])
  - Level Range: +10 ~ -40 dBm

```
>> Pow_com = set_RX_Ref_Level_ELSDR([0 -10; 1 -10])

Pow_com =

   -0.2450   -0.1785

>> P_rk1=10*log10(mean(abs(rk(1,:)).^2))-Pow_com(1)

P_rk1 =

   -9.2550

>> P_rk2=10*log10(mean(abs(rk(2,:)).^2))-Pow_com(2)

P_rk2 =

   -7.9660
```

- Reference level is the top edge of display and represents the maximum expected power at the YTPC1002C01 RX input.
- Pow_com is required for accurate dBm calculation.

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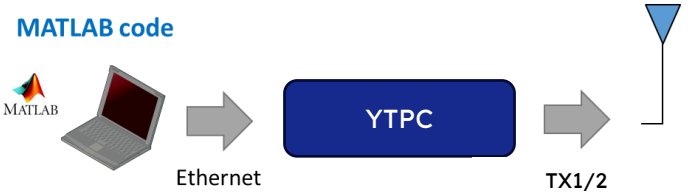
TX()



- TX(data, rep, time)
  - data: transmitted data
  - rep: number of repetitions (0 for infinite)
  - time (optional) : transmission period (seconds), only valid when rep equals to zero.

Case1: TX starts to transmit for once!

```
freq=1.92e6;
rep=0;  % infinite repeat
xk = single_tone_generator(freq);
TX(xk, 1);
```

Case2: TX starts to transmit for 10 seconds!

```
freq=1.92e6;
rep=0;  % infinite repeat
xk = single_tone_generator(freq);
TX(xk, rep,10);
```

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TX()

xk

$$\underbrace{\begin{bmatrix} 0.2 + i0.5 & \cdots & 0.9 - i0.1 \\ -0.3 + i0.7 & \cdots & -0.2 - i2 \end{bmatrix}}_{\text{Number of samples}} \left.\vphantom{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}\right\} \text{Dimension of RF chain}$$

- The range of the each real and imagery of each element of xk are $-1 \leq real\big(xk(m,n)\big) \leq 1$ and $-1 \leq imag\big(xk(m,n)\big) \leq 1$. ±1 means saturation.
- The time duration between $xk(m,n)$ and $xk(m,n+1)$ is the inverse of the sampling rate. In YTPC1002C01 the default sampling rate is 122.88MHz.
- *m*th row of *xk* corresponding to *m*th TX RF chain
- The limitation of total number of elements of rk is less than 8e7 (numel(xk) <= 8e7).
- Example code are in "./Demo_Code_for_YTPC1002C01/TX_only".
  - "TX_exp_with_TX_gain.m" and related function : simple example for illuminating how to transmit a single tone.

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TX_close()

- If use “Ctrl+C” to stop TX, this situation must use **TX_close** to close TX.

Ctrl+C

```
>> TX_exp_with_TX_gain
Operation terminated by user during TX (line 13)


In TX_exp_with_TX_gain (line 8)
TX(xk, rep,100);

>> TX_close

ans =

     1
```

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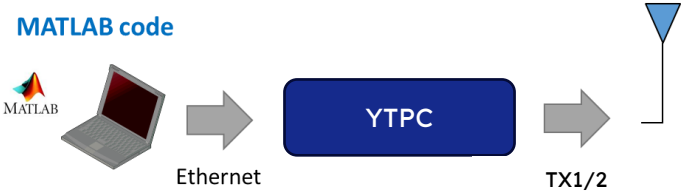
TX_start()



- TX_start(data, rep)
  - data: transmitted data
  - rep: 0, for infinite
- Use TX_start() to begin continuous transmission
  - Keeps transmitting until TX_close() is called
- Allows RX() to run during transmission
  - Enables bidirectional testing

Note: TX and RX must use the same number of RF chains, mismatched settings may cause instability

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set_TX_power()

- set_TX_power()
  - Input array: ([RF_chain0 Amp_val0; RF_chain1 Amp_val1])
- For example:
  - set TX power to -10 dBm
    - set_TX_power([0 -10; 1 -10])
  - Power control range: 50 dB

```
>> set_TX_power([0 -10; 1 -10]);
```

- This type is used to control the transmitted power by TX amplitude(dBm). This function will modify the TX power by using the calibration table.

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LO_Change()

- LO_Change(idx_SDR, tx_lo_1, rx_lo_1)
  - SDR default frequency: 1.0 GHz
  - Frequency range:
    - YTPC1002C01: 300MHz~6GHz, index SDR: 0
  - Unit: MHz
- This API is used for change center frequency of SDR. Three parameters are needed in this function, index of SDR, TX Frequency, RX Frequency.
  - For example:

LO_CHANGE(0, 1000,3000)

Change SDR0, TX1&TX2 Fc=1GHz, RX1&RX2 Fc= 3GHz,

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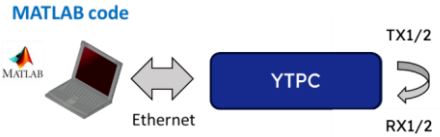
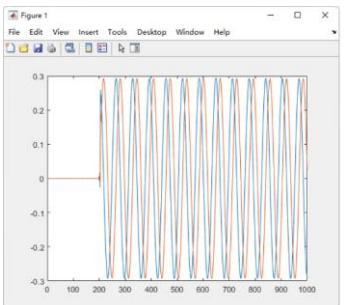
TX_RX()

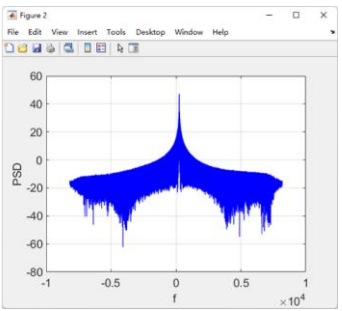
- TX_RX is for loopback, note that the relation between TX sample and RX sample is fixed for different runs and controlled by RX_delay
- TX_RX(tx_data, TX_REP, RX_LEN, RX_delay, LB_mode)
  - **Feature: TX+RX**
  - TX_REP: TX transmit data once or repeatedly, once: 0, repetition: 1
  - RX_LEN: RX log length
  - RX_delay: RX delayed reception length
  - LB_mode: 0: External loop back, 1: Internal loop back
- The example is in ".\ Demo_Code_for_YTPC1002C01 \TX_RX_delay"




```
TX_REP=0; % 0: infinite repeat;   1: once
RX_LEN=1000;
RX_delay=0;
LB_mode=0; % 0: RF; 1:memory loop back

rec_data=TX_RX(tx_data*0.8, TX_REP, RX_LEN, RX_delay, LB_mode);
```





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EXAMPLES

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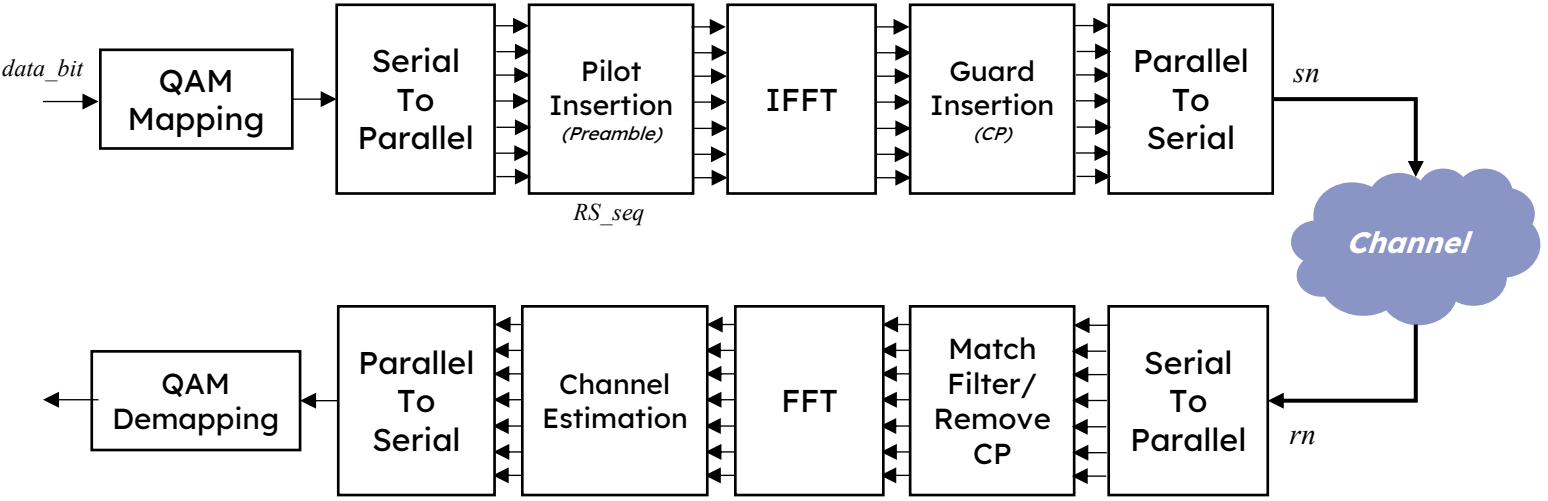
Structure of OFDM system



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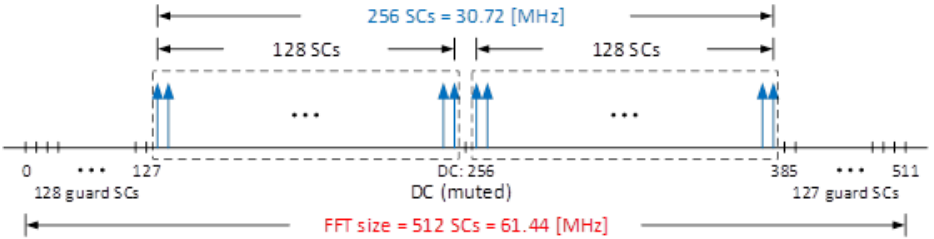
SISO OFDM - *Single Input Single Output*

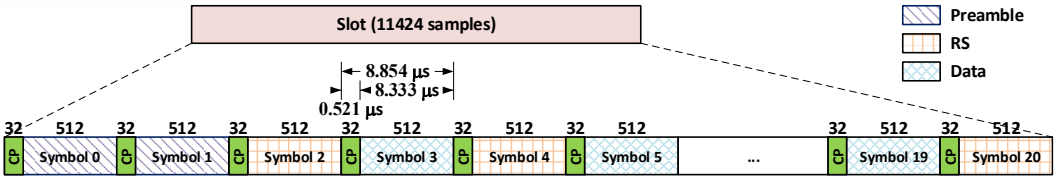
Parameters

- Carrier frequency: $f_c = 3$ GHz
- Sampling frequency: $F_s = 61.44$ MHz
- Sampling duration: $T_s = 1/F_s = 0.0163$ μs
- FFT size: $N_{FFT} = F_s/\Delta f = 512$ samples
- CP length: $N_{CP} = 32$ samples

Resource Bandwidth



Frame Structure



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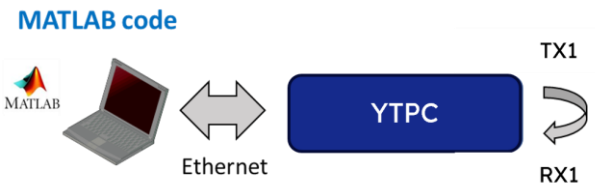
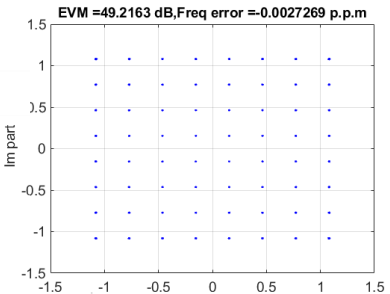
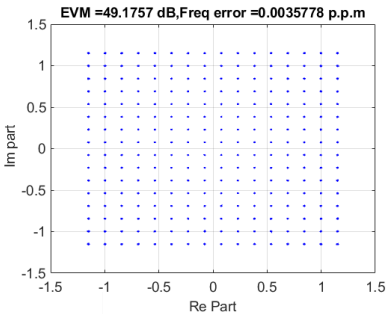
OFDM Loop Back Test

- Ensure that cables or antennas connected to the YTPC TX1-RX1.
- If user need to know the signal specifications, please refer to YTPC100x SISO_OFDM.pdf
- path:“ .\release\Demo_Code_for_YTPC1002C01\OFDM_EVM_Test_SISO\Test_EVM_TXLB .m “








```
N_cpc_data=(mod(idx_test,5)+1)*2;
C_init_data=1253233+idx_test_fixed*1000;
sn=TX_RJ_v4(Fs, Inx_Used_Carrier, total_num_sym, FFT_SIZE, CP_len, N_cpc_data, C_init_data, C_init_RS, preamble_seq_all);

RX_LEN=length(sn)+1000;
rn=TX_RX(sn*0.4, TX_REP, RX_LEN, RX_delay, LB_mode);
```

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SISO OFDM EVM Test with Coding

- Packet with random data Transmit/Receive
- Picture Transmit/Receive
- Webcam Transmit/Receive

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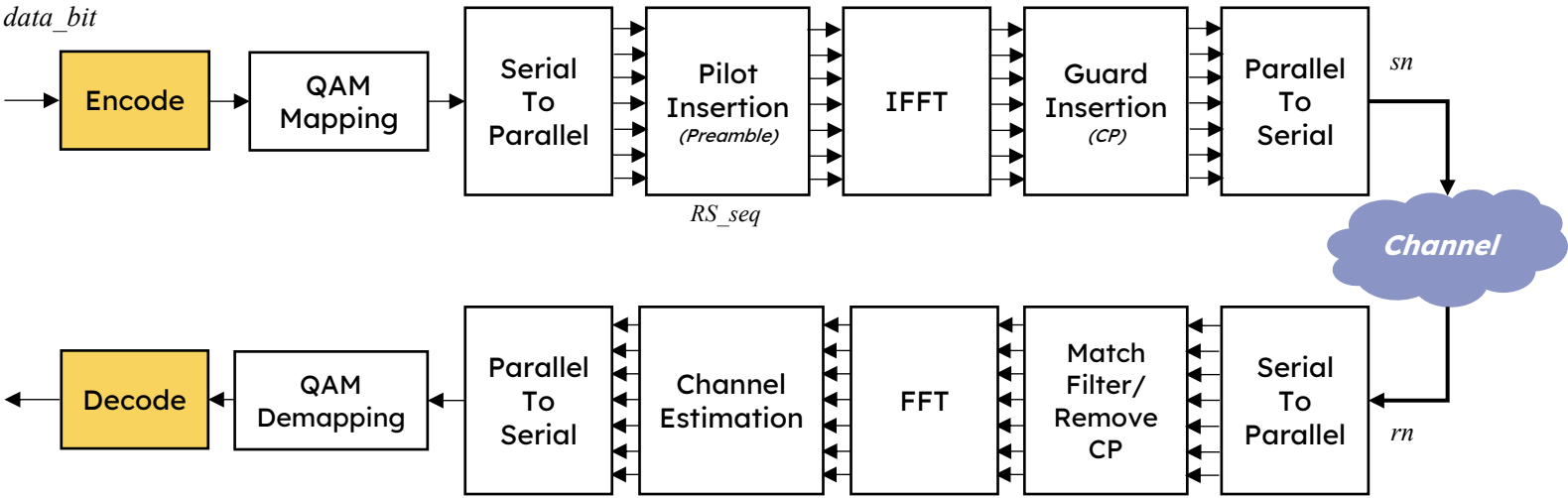
Structure of OFDM system with coding



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OFDM EVM Test with Coding

- ***Code Rate:***

$$\frac{k}{n}$$

k → every *k* bits of useful information

n → the coder generates a total of *n* bits of data

  - The code rate of a convolutional code will typically be ½, ⅔, ¾, ⅚, ⅞, etc.

- ***MCS Table:*** Modulation and Coding Scheme Index Table

MCS Index	Modulation	Code Rate	MCS Index	Modulation	Code Rate	MCS Index	Modulation	Code Rate
0	QPSK	1/2	4	16QAM	1/2	8	64QAM	2/3
1	QPSK	2/3	5	16QAM	2/3	9	64QAM	3/4
2	QPSK	3/4	6	16QAM	3/4	10	64QAM	5/6
3	QPSK	5/6	7	16QAM	5/6	If MCS_index=10, SNR should be > 23dB.		

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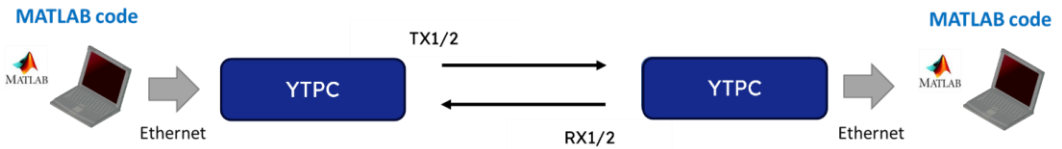
OFDM EVM Test with Coding (Transfer Pictures)

- Transpose data to picture based on OFDM_EVM_Test_SISO, and do the encoding processing.
- PC1(TX) path:
  " .\release\Demo_Code_for_YTPC1002C01\OFDM_EVM_with_coding\Picture_with_coding_BS.m "
- PC2(RX) path:
  " .\release\Demo_Code_for_YTPC1002C01\OFDM_EVM_with_coding\Picture_with_coding_UE.m "



BS

```
%=== open picture file and generate information bits ========
fid = fopen(strcat(name,'_', num2str(seq_image,'%05d'), '.jpg'), 'r');
A = fread(fid, 'uint8');
B = de2bi(A);
C = reshape(B', 1, numel(B));
BS.TX.info_ak=C;
seq_image=seq_image+1;
seq_image=mod(seq_image,seq_image_max)+1;
fclose(fid);
%==========================================================

sn=BS_TX_state14_v1(lnx_Used_Carrier, FFT_SIZE, CP_len,lnx_sym_preamble,preamble_seq);
TX(sn*0.4, 1);
```

UE

```
rn = RX(1,2000000*2);
CRC_result=UE_RX_ctrl_r2(rn,lnx_Used_Carrier, FFT_SIZE, CP_len,lnx_sym_preamble,preamble_seq);
```

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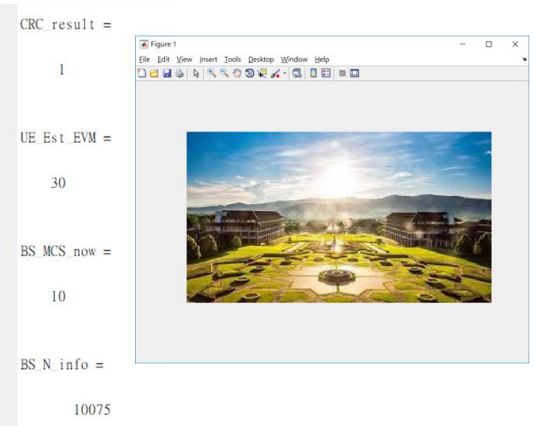
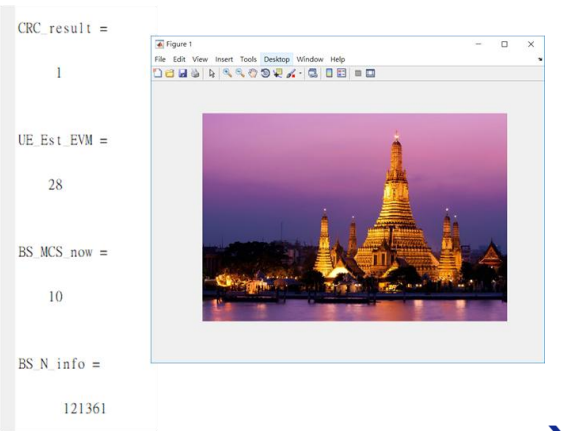
OFDM EVM Test with Coding (Transfer Pictures)

- Transpose data to picture based on OFDM_EVM_Test_SISO, and do the encoding processing.
- PC1(TX) path:
  " .\release\Demo_Code_for_YTPC1002C01\OFDM_EVM_with_coding\Picture_with_coding_BS.m "
- PC2(RX) path:
  " .\release\Demo_Code_for_YTPC1002C01\OFDM_EVM_with_coding\Picture_with_coding_UE.m "

Result for UE





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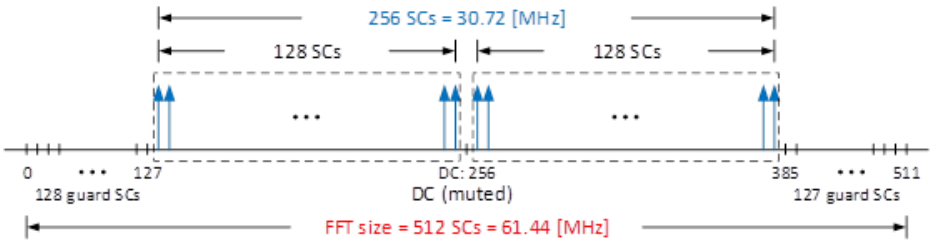
MIMO OFDM - *Multi-Input Multi-Output*

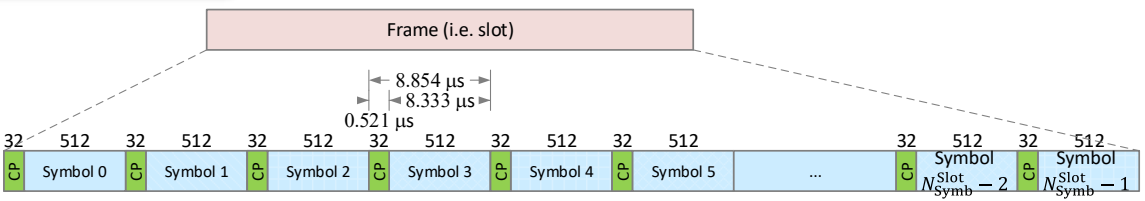
Parameters

- Carrier frequency: $f_c = 3$ GHz
- Sampling frequency: $F_s = 61.44$ MHz
- Sampling duration: $T_s = 1/F_s = 0.0163$ μs
- FFT size: $N_{FFT} = F_s/\Delta f = 512$ samples
- CP length: $N_{CP} = 32$ samples

Resource Bandwidth



Frame Structure



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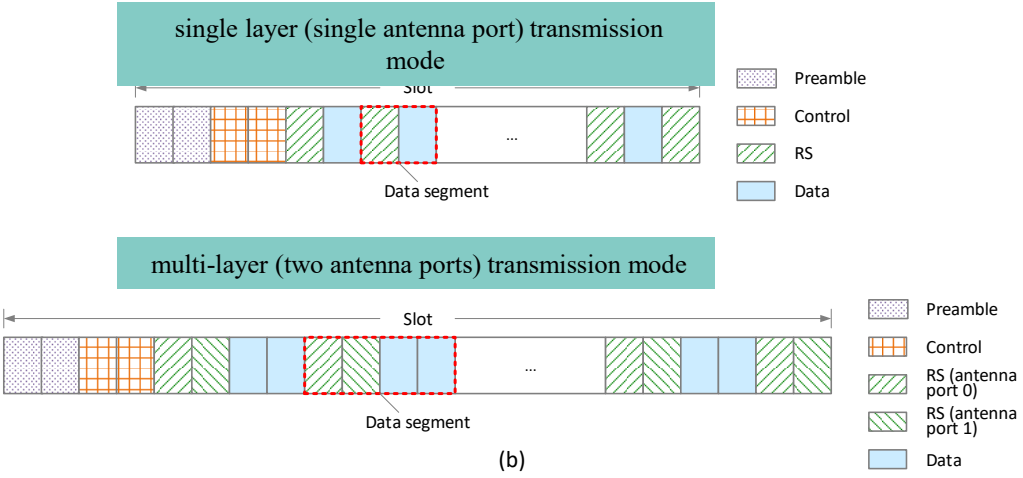
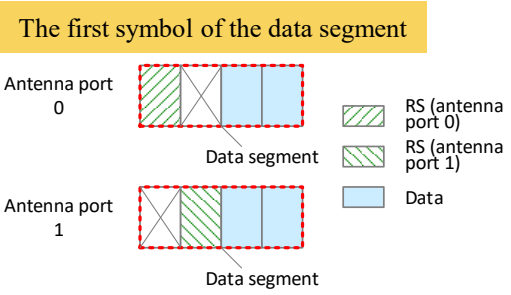
MIMO OFDM - *Multi-Input Multi-Output*

DL slot structure



(b)



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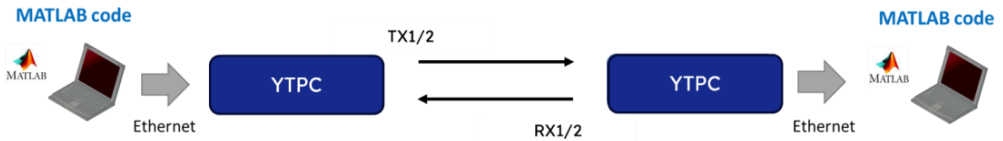
2x2 MIMO OFDM Test

- This example requires TRX1 and TRX2
- If user need to know the signal specifications, please refer to YTPC100 SDR_MIMO_OFDM.pdf
- PC1(TX) path: “ .\release\Demo_Code_for_YTPC1002C01\MIMO_OFDM_2x2_example\ Test_EVM_Tx_MIMO_2x2_two_layer_XP.m “
- PC2(RX) path: “ .\release\Demo_Code_for_YTPC1002C01\MIMO_OFDM_2x2_example\ Test_EVM_Rx_MIMO_2x2_two_layer_XP.m “

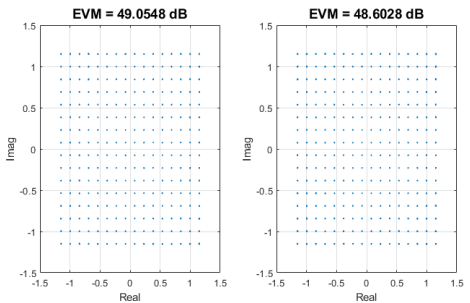


TX

```
Data_Parameter.N_cpc_Data = (mod(idx_test,5)+1)*2;

%=== MIMO-OFDM TX generation ===
sn = gen_MIMO_OFDM_TX_r1_XP(System_Parameter.Num_Layer, Data_Parameter.N_cpc_Data, Data_Parameter.C_init_Data, Preamble_Parameter.preamble_seq_all);
%==============================

% TX
TX(sn*0.4, 1);
```

RX

```
% RX
rn = RX(2,224128*2 + 1000);

[N_cpc_est, inx_start]=find_signal_header_CF(rn(1,:), System_Parameter.Inx_Used_Carrier,System_Parameter.FFT_SIZE,System_Parameter.CP_Len);
[rn_Equal, rn_Equal_Data, rn_Equal_Data1, rn_Equal_Data2, EVM, EVM1, EVM2, fe_est] = detect_MIMO_OFDM_RX_r2_XP(rn, 0, ...
                                                System_Parameter.Num_Layer,  Data_Parameter.C_init_Data, ...
                                                Preamble_Parameter.preamble_seq_all, N_cpc_est, inx_start);
```



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